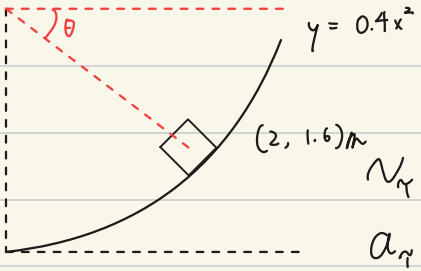
MIT 2.003SC PS2

Cyrus Hong (Awroncys)

Made with Goodnotes

Problem 1

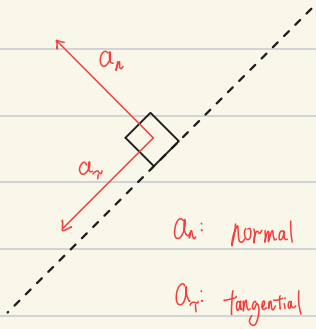
Pre-req: Radius of curvature: $\dfrac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$



$v_T = 8 ms^{-1}$

$a_T = 4 ms^{-2}$

fbd: instantaneous gradient



a_n: normal acceleration

a_T: tangential acceleration

a) find magnitude of acceleration & normal component

b) Draw fbd, determine friction coefficient

a) finding normal component

$\therefore a_n = \omega^2 R$, where v is tangential velocity

$= \dfrac{v^2}{R}$

Given $v_T = 8 ms^{-1}$

$y = 0.4x^2$

$\dfrac{dy}{dx} = 0.8x$

$\dfrac{d^2y}{dx^2} = 0.8$

$\rho = \dfrac{\left(1+\left(\frac{dy}{dx}\right)^2\right)^{3/2}}{\frac{d^2y}{dx^2}} = \dfrac{\left(1+(0.8x)^2\right)^{3/2}}{0.8}$

$\because (2, 1.6)m$, $x = 2$

$\rho \simeq 8.37$ (radius of curvature)

$a_n = \dfrac{v^2}{R} = \dfrac{8^2}{8.37} \approx 7.62 ms^{-2}$

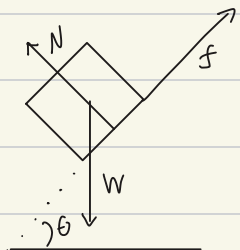
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Magnitude $= \sqrt{a_n^2 + a_T^2}$

$= \sqrt{7.62^2 + 4^2}$

$\approx 8.606\ ms^{-2}$

b) draw fbd



Find friction coefficient

knowing $a_y = 4\ ms^{-2}$

$$mg\sin\theta - \mu\left(m\frac{v^2}{R} + mg\cos\theta\right) = ma$$

$$\mu = \frac{g\sin\theta - a}{\frac{v^2}{12} + g\cos\theta}$$

$$\tan\theta = \frac{dy}{dx} = 0.8 \times 2$$

$$\theta \approx 57.995°$$

$$\mu = \frac{9.81\sin 57.995° - 4}{\frac{8^2}{8.37} + 9.81\cos 57.995°}$$

$\mu \approx 0.336$

Answer:

$\mu = 0.336$

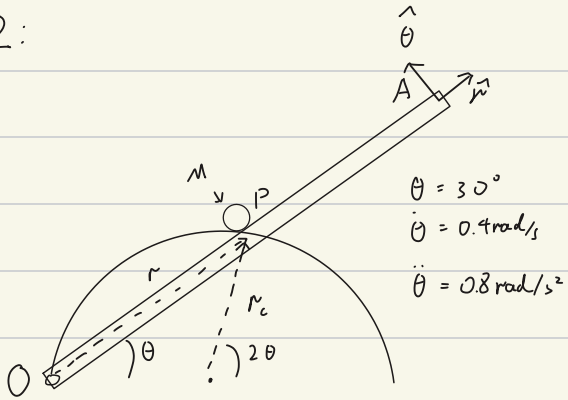
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Problem 2:

0.5 lb ball

$r_c = 0.4$ ft

$r_{P/O} = 2r_c \cos\theta$



$\theta = 30°$

$\dot{\theta} = 0.4$ rad/s

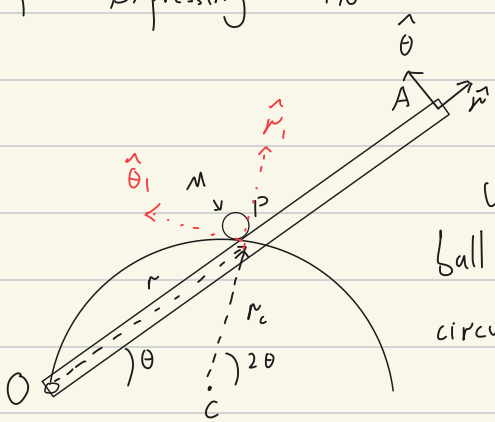
$\ddot{\theta} = 0.8$ rad/s^2

Find force of the arm on the ball

Step 1: Express $a_{P/O}$ in $\hat{r}$, $\hat{\theta}$ coords

Step 2: Use $\sum F = ma$ in $\hat{r}$, $\hat{\theta}$ directions.

Step 1: Expressing $a_{P/O}$



Use the information that the ball must traverse along the circular path

$$v_{P/O} = v_{P/C} + v_{C/O}$$

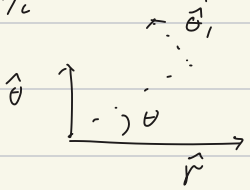
$$v_{P/C} = \left(\frac{\partial r_{P/C}}{\partial t}\right)_{\dot{\theta}=0} + \vec{\omega} \times r_{P/C}$$

$$= 0 + 2\dot{\theta}\hat{k} \times (r_c \hat{r}_1)$$

$$= 2\dot{\theta} r_c \hat{\theta}_1 \;//$$

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$$v_{p/c} = 2r_c\dot{\theta}\hat{\theta}_1$$



$$\hat{\theta}_1 = -\sin\theta\hat{r} + \cos\theta\hat{\theta}$$

$$v_{p/c} = 2r_c\dot{\theta}\left(-\sin\theta\hat{r} + \cos\theta\hat{\theta}\right)$$

$$a_{p/c} = -2r_c\ddot{\theta}\sin\theta\hat{r} - 2r_c\dot{\theta}^2\cos\theta\hat{r} + 2r_c\ddot{\theta}\cos\theta\hat{\theta} - 2r_c\dot{\theta}^2\sin\theta\hat{\theta}$$

$$= \left[-2r_c\dot{\theta}^2\cos\theta - 2r_c\ddot{\theta}\sin\theta\right]\hat{r} + \left[2r_c\ddot{\theta}\cos\theta - 2r_c\dot{\theta}^2\sin\theta\right]\hat{\theta}$$

$\therefore a_{p/o} = a_{p/c} + a_{c/o}$ and $a_{c/o} = 0$

$$a_{p/o} = \left[-2r_c\dot{\theta}^2\cos\theta - 2r_c\ddot{\theta}\sin\theta\right]\hat{r} + \left[2r_c\ddot{\theta}\cos\theta - 2r_c\dot{\theta}^2\sin\theta\right]\hat{\theta}$$

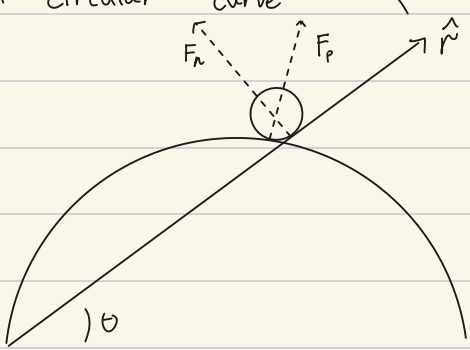
Step 2. find forces and apply $F = ma$

F_N = comes from rod

F_p = from circular curve

$$\sum_{\hat{r}} F = F_p\cos\theta - mg\sin\theta = m a_{\hat{r}}$$

$$\sum_{\hat{\theta}} F = F_p\sin\theta + F_N - mg\cos\theta = m a_{\hat{\theta}}$$



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Solving $\sum_{\hat{r}} F$

$$\sum_{\hat{r}} F = F_p \cos\theta - mg\sin\theta = ma_{\hat{r}}$$

$$a_{\hat{r}, P/O} = \left[-2r_c\dot{\theta}^2\cos\theta - 2r_c\ddot{\theta}\sin\theta\right]\hat{r}$$

$r_c = 0.4$ ft

$\theta = 30^\circ$

$\dot{\theta} = 0.4$ rad/s

$\ddot{\theta} = 0.8$ rad/s

$$a_{\hat{r}, P/O} \approx -0.43085 \text{ ft/s}^2\,\hat{r}$$

convert 0.5 lb to slugs

$$\frac{0.5}{32.2} \approx 0.0155$$

$$F_p\cos\theta - mg\sin\theta = m(-0.43085)$$

$$F_p \approx 0.281 \text{ lbs}$$

Solving $\sum_{\hat{\theta}} F$

$$\sum_{\hat{\theta}} F = F_p\sin\theta + F_n - mg\cos\theta = ma_{\hat{\theta}}$$

$$a_{\hat{\theta}} = 2r_c\dot{\theta}\dot{\theta}\cos\theta - 2r_c\dot{\theta}^2\sin\theta$$

$$= 2(0.4)(0.8)\cos 30^\circ - 2(0.4)(0.4)^2\sin 30^\circ$$

$$\approx 0.4903 \text{ ft s}^{-2}$$

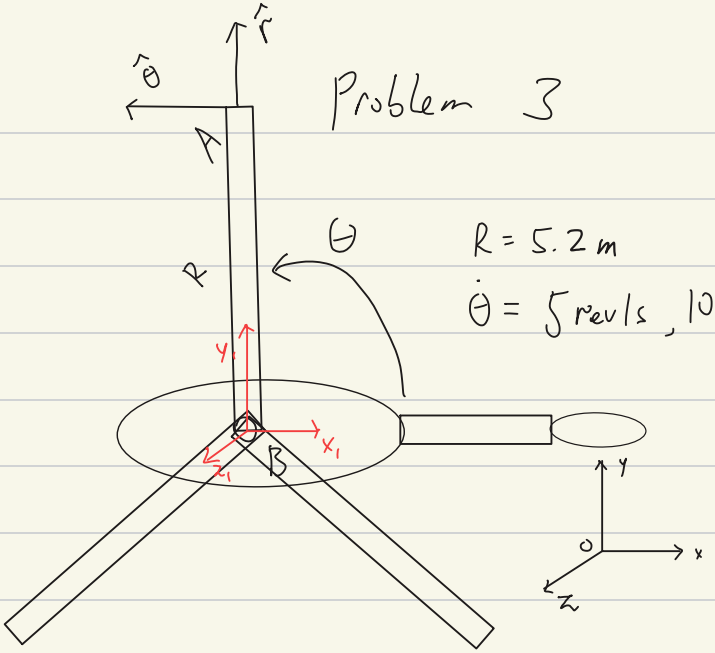
$$F_p\sin\theta + F_n - mg\cos\theta = m\hat{a}_\theta$$

With calculated F_p value:

$$F_n \approx 0.300 \text{ lbs}$$

Made with Goodnotes

Problem 3



$R = 5.2\ m$

$\dot{\theta} = 5\ rev/s,\ 10\pi\ rad/s$

flying in a st line

5 m/s velocity

$0.5\ m/s^2$ acceleration

$B_{x,y,z}$ attached to helicopter

O_{xyz} is inertial ref. frame

Find $v_{A/O}$, $a_{A/O}$ when $\theta = 90^\circ$

Step 1: Express $r_{A/O}$

Step 2: Keep taking derivatives.

$\hat{r} = \cos\theta\,\hat{i}_1 + \sin\theta\,\hat{j}_1$

$\dot{\hat{r}} = -\sin\theta\,\dot{\theta}\,\hat{i}_1 + \cos\theta\,\dot{\theta}\,\hat{j}_1$

Step 1: Expressing $r_{A/O}$

$r_{A/O} = r_{A/B} + r_{B/O}$

$v_{A/O} = \left(\frac{\partial r_{A/B}}{\partial t}\right)_{\omega=0} + \vec{\omega} \times r_{A/B} + v_{B/O}$

$= 0 + \dot{\theta}\hat{k} \times R\hat{r} + v_{B/O}$

$= \dot{\theta} R \hat{\theta} - 5\hat{i}$

$= \dot{\theta} R(-\hat{i}\sin\theta + \hat{j}\cos\theta) - 5\hat{i}$

($B_{x,y,z}$ aligned with O_{xyz})

Made with Goodnotes

$$v_{A/O} = \dot{\theta} R(-\sin\theta\,\hat{i} + \cos\theta\,\hat{j}) - 5\hat{i}$$

$$= (-\dot{\theta} R\sin\theta - 5)\hat{i} + \dot{\theta} R\cos\theta\,\hat{j}$$

$\theta = 90^\circ$, $\dot{\theta} = 10\pi$ rad/s, $R = 5.2$ m

$$v_{A/O} = (-52\pi - 5)\ \text{m/s}\ \hat{i}$$

Finding $a_{A/O}$

Proof $\frac{d}{dt}(\hat{r}) = \dot{\theta}\hat{\theta}$

$\frac{d}{dt}(\cos\theta\,\hat{i} + \sin\theta\,\hat{j}) = -\sin\theta\,\dot{\theta}\hat{i} + \cos\theta\,\dot{\theta}\hat{j}$

$= \dot{\theta}(-\sin\theta\,\hat{i} + \cos\theta\,\hat{j})$

$= \dot{\theta}\hat{\theta}$

$$v_{A/O} = v_{B/O} + v_{A/B}$$

$$= v_{B/O} + \frac{d}{dt}(r\hat{r})$$

$$= v_{B/O} + \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} \quad \text{proved}$$

$$a_{A/O} = a_{B/O} + \ddot{r}\hat{r} + \dot{r}\dot{\theta}\hat{\theta} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} - r\dot{\theta}^2\hat{r} \quad \text{proved}$$

$$= a_{B/O} + (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

Proof $\frac{d}{dt}(\hat{\theta}) = -\dot{\theta}\hat{r}$

$\frac{d}{dt}(-\sin\theta\,\hat{i} + \cos\theta\,\hat{j}) = -\cos\theta\,\dot{\theta}\hat{i} - \sin\theta\,\dot{\theta}\hat{j}$

$= -\dot{\theta}\hat{r}$

$\because \ddot{r} = \dot{r} = \ddot{\theta} = 0$

$$a_{A/O} = -0.5\hat{i} - r\dot{\theta}^2\hat{r}$$

$$a_{A/O} = -0.5\hat{i} - r\dot{\theta}^2(\cos\theta\,\hat{i} + \sin\theta\,\hat{j}) \quad (B_{x_1y_1z_1} \text{ aligned with } O_{xyz} \text{ when } \theta = 90^\circ)$$

$$= (-0.5 - r\dot{\theta}^2\cos\theta)\hat{i} - r\dot{\theta}^2\sin\theta\,\hat{j}$$

$\dot{\theta} = 10\pi$ rad/s, $\theta = 90^\circ$, $r = 5.2$ m

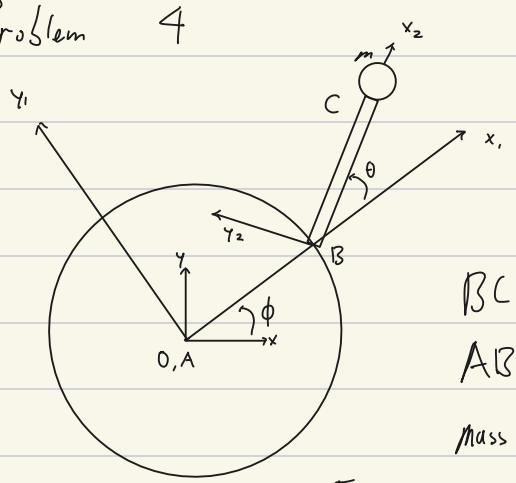
$$= -0.5\hat{i} - 520\pi^2\hat{j}$$

Answer

$v_{A/O} = (-52\pi - 5)\hat{i}$, $a_{A/O} = -0.5\hat{i} - 520\pi^2\hat{j}$

Made with Goodnotes

Problem 4



$\dot{\theta}$ = 10 rev/min = 20π rad/min = $\frac{1}{3}\pi$ rad/s

$\dot{\phi}$ = 6 rev/min = 12π rad/min = $\frac{1}{5}\pi$ rad/s

both counter clockwise, in $\hat{k}$ direction.

BC length = 2m

AB length = 6m

Mass passenger = 75kg

Find $v_{C/O}$, $a_{C/O}$

Step 1: Express $r_{C/O}$

Step 2: Take derivatives

(Just treat it like the last P)

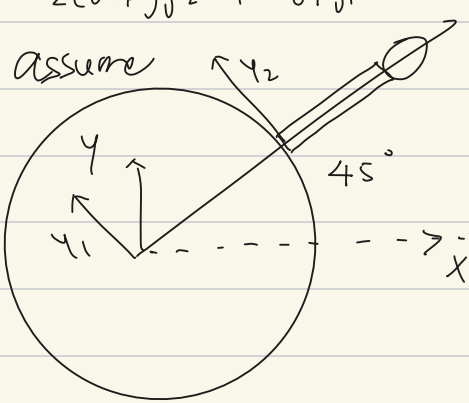
Expressing $r_{C/O}$: $r_{C/O} = r_{C/B} + r_{B/A} + r_{A/O}$

Start taking time derivatives

$$v_{C/O} = \left(\frac{\partial r_{C/B}}{\partial t}\right)_{w=0} + \vec{\omega} \times r_{C/B} + \left(\frac{\partial r_{B/A}}{\partial t}\right)_{w=0} + \vec{\omega} \times r_{B/A} + \left(\frac{\partial r_{A/O}}{\partial t}\right)_{w=0} + \vec{\omega} \times r_{A/O}$$

$$= 0 + (\dot{\theta}+\dot{\phi})\hat{k} \times 2\hat{i}_2 + 0 + \dot{\phi}\hat{k} \times 6\hat{i}_1 + 0$$

$$= 2(\dot{\theta}+\dot{\phi})\hat{j}_2 + 6\dot{\phi}\hat{j}_1$$



now, $\hat{j}_1 = \hat{j}_2 = -\frac{\sqrt{2}}{2}\hat{i} + \frac{\sqrt{2}}{2}\hat{j}$

$$v_{C/O} = (2\dot{\theta} + 8\dot{\phi})\left(-\frac{\sqrt{2}}{2}\hat{i} + \frac{\sqrt{2}}{2}\hat{j}\right)$$

Made with Goodnotes

Finding $a_{C/O}$

$$\Lambda_{C/O} = \Lambda_{C/B} + \Lambda_{B/O}$$

$$\Lambda_{B/O} = \Lambda_{B/A} + \Lambda_{A/O}$$

$$v_{B/O} = v_{B/A,\,w=0} + \vec{\omega} \times \Lambda_{B/A} + v_{A/O}$$

$$= a_{B/A} + a_{A/O} + \dot{\vec{\omega}} \times \Lambda_{B/A} + \vec{\omega} \times v_{B/A} + \dot{\vec{\omega}} \times \Lambda_{B/A} + \vec{\omega} \times v_{B/A} + \vec{\omega} \times (\vec{\omega} \times \Lambda_{B/A})$$

$$= a_{B/A} + a_{A/O} + \vec{\alpha} \times \Lambda_{B/A} + \vec{\omega} \times (\vec{\omega} \times \Lambda_{B/A}) + 2\vec{\omega} \times v_{B/A}$$

Finding $\frac{d}{dt}(v_{B/A})$

$$= \left(\frac{\partial v_{B/A}}{\partial t}\right)_{\omega=0} + \vec{\omega} \times v_{B/A}$$

$$= a_{B/A} + \vec{\omega} \times v_{B/A}$$

Finding $\frac{d}{dt}(\vec{\omega} \times \Lambda_{B/A})$

$$= \dot{\vec{\omega}} \times \Lambda_{B/A} + \vec{\omega} \times \left(\left(\frac{\partial \Lambda_{B/A}}{\partial t}\right)_{\omega=0} + \vec{\omega} \times \Lambda_{B/A}\right) = \dot{\vec{\omega}} \times \Lambda_{B/A} + \vec{\omega} \times v_{B/A} + \vec{\omega} \times (\vec{\omega} \times \Lambda_{B/A})$$

$$a_{A/O} = a_{B/A} = \dot{\vec{\omega}} = v_{B/A}{}_{\omega=0} = 0$$

$$a_{B/O} = \dot{\phi}\hat{k} \times (\dot{\phi}\hat{k} \times 6\hat{i}_1) = -\dot{\phi}^2 6\hat{i}_1$$

finding $a_{C/B}$

$$\Lambda_{C/B} = \Lambda_{C/B}$$

$$v_{C/B} = \left(\frac{\partial \Lambda_{C/B}}{\partial t}\right)_{\omega=0} + \vec{\omega} \times \Lambda_{C/B}$$

$$a_{C/B} = a_{C/B}{}_{\omega=0} + \vec{\omega} \times v_{C/B} + \dot{\vec{\omega}} \times \Lambda_{C/B} + \vec{\omega} \times (\vec{\omega} \times \Lambda_{C/B}) + \vec{\omega} \times v_{C/B}$$

$$a_{C/B} = a_{C/B}{}_{\omega=0} + \dot{\vec{\omega}} \times \Lambda_{C/B} + \vec{\omega} \times (\vec{\omega} \times \Lambda_{C/B}) + 2\vec{\omega} \times v_{C/B}$$

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$$a_{C/B} = a_{C/B_{\omega=0}} + \dot{\vec{\omega}} \times r_{C/B} + \vec{\omega} \times (\vec{\omega} \times r_{C/B}) + 2\vec{\omega} \times v_{C/B}$$

$$a_{C/B_{\omega=0}} = \dot{\vec{\omega}} = v_{C/B} = 0$$

$$a_{C/B} = \vec{\omega} \times (\vec{\omega} \times r_{C/B})$$

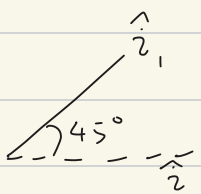
$$= (\dot{\theta} + \dot{\phi})\hat{k} \times ((\dot{\theta} + \dot{\phi})\hat{k} \times 2\hat{i}_2)$$

$$= -2(\dot{\theta} + \dot{\phi})^2 \hat{i}_1$$

$$a_{C/O} = a_{C/B} + a_{B/O}$$

$$= -2(\dot{\theta} + \dot{\phi})^2 \hat{i}_1 + (-\dot{\phi}^2 6 \hat{i}_1)$$

$$= (-2(\dot{\theta} + \dot{\phi})^2 - 6\dot{\phi}^2)\hat{i}_1$$



$$\hat{i}_1 = \cos 45^\circ \hat{i} + \sin 45^\circ \hat{j}$$

$$= \frac{\sqrt{2}}{2}\hat{i} + \frac{\sqrt{2}}{2}\hat{j}$$

$$a_{C/O} = (-2(\dot{\theta} + \dot{\phi})^2 - 6\dot{\phi}^2)\left(\frac{\sqrt{2}}{2}\hat{i} + \frac{\sqrt{2}}{2}\hat{j}\right)$$

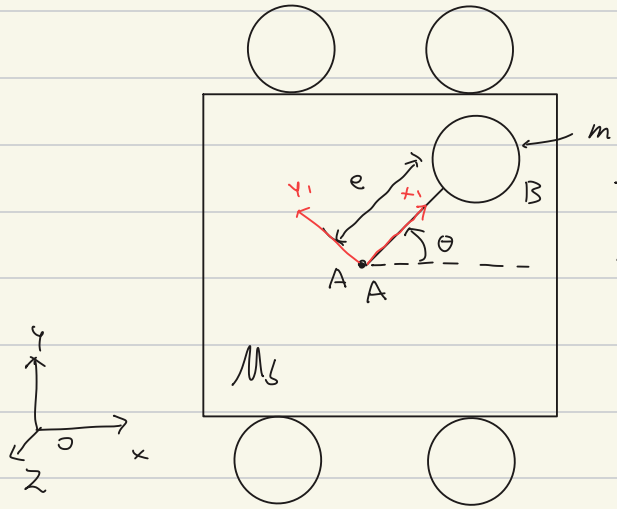
Answer: under assumption $\hat{i}_2 = \hat{i}_1$ and $\hat{i}_1 = \cos 45^\circ \hat{i} + \sin 45^\circ \hat{j}$

$$v_{C/O} = (2\dot{\theta} + 8\dot{\phi})\left(-\frac{\sqrt{2}}{2}\hat{i} + \frac{\sqrt{2}}{2}\hat{j}\right)$$

$$a_{C/O} = (-2(\dot{\theta} + \dot{\phi})^2 - 6\dot{\phi}^2)\left(\frac{\sqrt{2}}{2}\hat{i} + \frac{\sqrt{2}}{2}\hat{j}\right)$$

Made with Goodnotes

Problem 5



Let $a_{A/O} = \ddot{x}\hat{i}$

$\vec{\omega} = \dot{\theta}\hat{k}$

- Find $a_{B/O}$
- Find magnitude & direction of the force the rod supplies to the mass.
- Find magnitude & direction of the force the rod exerts to A.

Finding $a_{B/O}$

$$r_{B/O} = r_{B/A} + r_{A/O}$$

$$v_{B/O} = v_{B/A_{A_{x_1y_1z_1}}} + \vec{\omega} \times r_{B/A} + v_{A/O}$$

$$a_{B/O} = a_{B/A_{A_{x_1y_1z_1}}} + \vec{\omega} \times (\vec{\omega} \times r_{B/A}) + a_{A/O} + 2\vec{\omega} \times v_{B/A}$$

$$a_{B/A_{A_{x_1y_1z_1}}} = v_{B/A_{A_{x_1y_1z_1}}} = 0$$

$$a_{B/O} = -e\dot{\theta}^2 \hat{i}_1 + \ddot{x}\hat{i}$$

$$\hat{i}_1 = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}$$

$$a_{B/O} = (-e\dot{\theta}^2\cos\theta + \ddot{x})\hat{i} + (-e\dot{\theta}^2\sin\theta)\hat{j}$$

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force on B by rod

$\because$ we know $a_{B/O}$

$$\sum F = m a_{B/O} = m\left[\left(\ddot{x} - e\dot{\theta}^2\cos\theta\right)\hat{i} - e\dot{\theta}^2\sin\theta\,\hat{j}\right] = F_{rod} - mg\,\hat{j}$$

$$F_{rod} = m\left[\left(\ddot{x} - e\dot{\theta}^2\cos\theta\right)\hat{i} + \left(g - e\dot{\theta}^2\sin\theta\right)\hat{j}\right]$$

force on A by rod

$$F_{\substack{rod\\ on\\ m}} = -F_{\substack{rod\\ on\\ m}} \quad (N3\ Law)$$

$$F_{\substack{rod\\ on\\ A}} = m\left[\left(e\dot{\theta}^2\cos\theta - \ddot{x}\right)\hat{i} + \left(e\dot{\theta}^2\sin\theta - g\right)\hat{j}\right]$$

Answer:

$$a_{B/O} = \left(\ddot{x} - e\dot{\theta}^2\cos\theta\right)\hat{i} + \left(-e\dot{\theta}^2\sin\theta\right)\hat{j}$$

$$F_{rod\ on\ m} = m\left[\left(\ddot{x} - e\dot{\theta}^2\cos\theta\right)\hat{i} + \left(g - e\dot{\theta}^2\sin\theta\right)\hat{j}\right]$$

$$F_{rod\ on\ A} = m\left[\left(e\dot{\theta}^2\cos\theta - \ddot{x}\right)\hat{i} + \left(e\dot{\theta}^2\sin\theta - g\right)\hat{j}\right]$$

Made with Goodnotes